The mathematical destination

Finite-horizon stopping theorem. Fix the number of offers before a round. Every offer is an independent draw from the same known finite distribution of scores. Take the current score and end, or pass it permanently; the final offer must be taken, including zero. Only the accepted score counts. There are no fees or recall. The objective is the *expected accepted score*: the probability-weighted average over possible rounds.

Let V_m be the best average *before seeing* any of m remaining offers. If X is one fresh offer, then

$$V_1 = \mathbb{E}[X], \qquad V_m = \mathbb{E}[\max\{X, V_{m-1}\}] \quad (m > 1).$$

With $m > 1$ offers left *including the current offer* x, take when $x > V_{m-1}$ and pass when $x < V_{m-1}$. At equality either choice is optimal. The last offer has no pass option. The recurrence compares taking with the best *whole remaining plan*, rather than just the next draw.

Replacement and independence make every passed history irrelevant to the future distribution. Backward induction bounds every legal plan, even one that remembers earlier offers or randomizes its decisions. For tickets $0, 4, 6$,

$$V_1 = 10/3, \qquad V_2 = 40/9, \qquad V_3 = 134/27.$$

Thus take a current 4 with two offers left, but pass it with three. For $0, 3, 6$ with two offers, taking or passing 3 ties; both optimal plans total 36 on nine cases. For $0, 2, 6$ the middle 2 is below $V_1 = 8/3$ and is passed: total 34. More offers never lower the optimum. For any nonconstant finite bag they strictly raise it at each finite horizon; a constant-score bag gives equality. Proofs appear on pages 6–8.

Limits. A known bag without replacement requires its remaining inventory as part of the state. Recall, changing probabilities, fees, an unknown horizon, or an objective such as selecting the overall largest offer require a different model. This is expected-score stopping, not the rank-only secretary problem. The mathematical iid assumption is stronger than rolling a stickered die or returning a ticket: the drawing procedure needs rehearsal.

Experiments, conjectures and explanations. Six played rounds may suggest a rule. All equally likely full words evaluate a fixed rule exactly. A complete policy audit or backward argument establishes optimality. A favorable sample does not certify a better average. An expected score is not a promised score in the next round or short batch.

Actual student map: one eight-page packet, W60-S-v3.

Student pages	Band and route	What children can discover
1–3, 6–7	Grades 3–5; concrete core	Invent/apply rules; construct opposite samples; pool nine/27 cases; compare totals; find a tie, and a bag whose middle ticket is passed.
4–5 and 8	Grades 4–5; continuation	Compare continuation averages; prove a bound for history-dependent plans; change horizons; explain extra offers.
1, 2 and 6	K–1, adult-led (headed K–5)	Play legal two-offer rounds; compare take and pass cube towers row by row; see equal towers (guide p. 2).

Adult reading and addition can support the core. Children must compare scores, count turns, retain take/pass and a deadline, and apply a fixed rule. Upper work needs averages, fractional comparisons and readiness for a branching argument. Success can be one sound investigation; the packet can support several visits.

Prepare for eleven children and three anchored adults

KK11: two kindergartners and two first graders with the parent volunteer. **3333:** four third graders with a mathematician, in two pairs. **445:** two fourth graders and one fifth grader with the organizer; a pair and a rotating referee. Adults stay at their tables. This staffing is part of the proposed plan, not a tested independence claim.

Four playing kits: three older-table kits and one for KK11. Per kit: one blank die stickered $0, 0, 4, 4, 6, 6$ and labelled "bag $0, 4, 6$"; four counters; pencil and eraser. Each score is on two faces, so each has chance $1/3$, and every roll is a fresh draw from the bag with nothing to return or mix. Keep a spare die and four spare counters. The partner rolls; the face stays up while the player decides. Taking ends the round; after a pass the next roll is the next offer, and the partner prevents reclaiming passed offers. Check whether one counter remains *while deciding*; remove a counter only *after* the decision. *Fallback without dice:* three indistinguishable cardstock tickets $0, 4, 6$ in an opaque bag per kit, with an erasable offer display; write the offer on it, then return the ticket and mix.

Problems 7 and 8 use the printed cards, so new bags need no dice. To play a new bag, sticker a separate labelled die (faces $0, 0, 3, 3, 6, 6$ for $0, 3, 6$); never mix bags. For the launch, have practice tickets $1, 2, 5$. For the optional average example, have nine cubes and three sharing places. KK11 also needs 30 linking cubes for towers.

Print single-sided US Letter at 100%. Give one page at a time.

Print	Allocation	Sheets
Student page 1	Seven older children, the KK11 adult and one spare	9
Pages 2, 3, 6, 7 on paper	One working sheet per older kit, plus one spare of each page	16
Pages 2 and 6 on paper	KK11's tower pages, used uncut	2
Pages 4, 5, 8 on paper	Three upper children plus one spare of each; keep for readiness/later	12
Pages 2, 3, 6, 7 on cardstock	One complete card set for each older table: two copies of each	8
Spare cardstock	One spare copy of pages 2, 3, 6 and 7; leave uncut until needed	4
Adult guide	One per adult	3 guides

Each older table's set is nine $0/4/6$ pairs, 27 $0/4/6$ triples and nine pairs for each of $0/3/6$, $0/5/6$ and $0/2/6$: 63 cards. **Cutting is optional.** A sheet's rows already run by first offer, so cards can be scored in place; cut along dashed borders only when children want to move and sort them, and keep cut cards in envelopes labelled by bag and word length. Children share and partition a table's set; each child need not score all 27 cards. Cutting all 126 cards would take roughly 15–25 minutes; this estimate is untested.

K–1 route at KK11, adult-led, on pages 1, 2 and 6. The adult reads and records; children roll, decide, move counters and build towers. (1) After the shared launch, play two-offer rounds with the die and two counters; record six scores in Problem 1's two-offer row. (2) Problem 3, one first-offer row at a time: build a *take* cube tower from the row's first numbers and a *pass* tower from its second numbers, then compare heights. (3) The same with page 6's $0, 3, 6$ middle row. That K–1 children can compare towers this way is a prediction. **Plain-words key.** Any rule is legal, and six rounds cannot show which rule is better. First 0: take 0, pass $0 + 4 + 6 = 10$, so pass. First 4: take $4 + 4 + 4 = 12$, pass 10, so take. First 6: take 18, pass 10, so take. Bag $0, 3, 6$, first 3: take $3 + 3 + 3 = 9$, pass $0 + 3 + 6 = 9$; equal towers, so both choices are best. Week 1's K–1 tiling pages 1–2 stay available with the Week 1 guide.

A flexible hour and the handling pretest

Minutes	Proposed route
0–5	Movement/running around.
5–7	Handle dice, counters and cards freely.
7–10	Brief shared legal-round demonstration below, KK11 included.
10–25	Older pairs play Problem 1, swap roles, and construct Problem 2 examples. KK11 plays two-offer rounds with its adult.
25–40	Pool Problem 3's nine cards, then start Problem 4's 27 cards if ready; KK11 builds Problem 3's towers row by row. Save partial work and labelled totals.
40–43	Movement/reset; rotate player, dealer and referee.
43–55	Continue the same investigation or offer pages 6–7 (Problem 7); KK11 may try page 6's $0, 3, 6$ middle row. Upper children may choose page 4 or 5. Page 8 can wait for another visit.
55–60	Share one checked claim, then tidy.

Say and show a legal round, not the target strategy. Show the die: "Each number is on two faces, so this die is the bag $0, 4, 6$." Then show practice tickets $1, 2, 5$ and two counters. Say: "We set two offers. This number is the current offer. Take ends the round; pass gives it up." Arrange the practice offers 2 then 1 openly as a *demonstration*, not a fair draw. A child passes 2; the partner removes one counter after that decision. Show 1: "One counter is left now, so take this offer, even though it is smaller." The child records score 1. Ask one child to repeat when the counter is removed. Do not supply a taking rule for $0, 4, 6$.

The tables now set their horizon and put out one counter per offer before any roll. At 3333, two pairs alternate player/dealer. At 445, player/dealer/referee rotate; the referee checks the deadline and no recall. For cards, a reader, decision-maker and recorder can rotate. The adult may read or add while children retain the choice and check each word.

Fallbacks. KK11: keep playing legal two-offer rounds, or build towers for one first-offer row only. 3333: replay legal two-offer rounds with an existing child-invented rule, or use only one first-offer group of three cards, then pool it. 445: stage different possible rounds for two fixed rules and find a case favoring each; return to exact comparison when ready. Do not interpret an illegal round as evidence about a strategy.

Pretest with the actual materials before use. All remain unperformed.

- Roll each stickered die about 30 times: every score should turn up and the stickers should hold. A short sample cannot prove fairness. With the ticket fallback, check that tickets cannot be told apart, and rehearse the display, return and mixing.
- Rehearse passes that cannot be reclaimed, early taking that ends immediately, and a first 0 that may be passed with two counters. Rehearse compulsory final 0 and *after-decision* counter removal.
- Score one complete set of Problem 4 cards on both plans' lines, cut and uncut. Check legibility and pencil pressure. Retain full words and unseen tails throughout.
- Rehearse role rotation, four simultaneous kits with anchored adults, children retaining decisions, and building towers from a card row. Rolling, scoring, enforcement, staffing and classroom fit are unpiloted.

Closing question: "Which claim did we try, and which did we check on every possible card?" Finishing all pages is not the aim.

Concrete core / student pages 1–2

Problem 1 / Invent and play fixed rules; core

Let children choose two-offer and three-offer rules. A rule must say what to do when a nonfinal offer appears; it can depend on turns left or earlier passed offers, but cannot see unseen offers. Final acceptance is compulsory. Keep each rule fixed for its six rounds, then swap roles. Record only the six accepted scores on the worksheet and add afterward. If a disputed round needs review, keep one offer word with the accepted position circled; do not add concurrent tallies of draws, passes or successes.

There is no single required invented rule or six-round total. Legal examples include taking the first offer or passing until the last. The optimal rules are answers for later investigations, not a launch instruction. A playing sample can support a conjecture; it cannot certify the best average. **Hints, in order.** (1) "What will your rule do if the first offer is 0?" (2) Point to the counters: "Would that choice change with one counter left?" (3) Rehearse a rule on a pretend offer before another random round.

Problem 2 / Opposite six-round winners with unchanged rules; core

Use the children's same two fixed rules in both matches. Here is a complete possible construction; every shown word is in drawing order, including any unseen suffix. Each row uses the displayed word in *all six rounds*.

Match	Two-offer rounds	Three-offer rounds	Six-round totals
I	$6 \times (6, 6)$	$6 \times (0, 0, 0)$	36 versus 0
II	$6 \times (0, 0)$	$6 \times (6, 6, 6)$	0 versus 36

Any legal rule scores the common value on a constant word, because it must eventually take an offer. Thus these examples work without changing either rule, whatever the children invented. Every full word has positive probability. These are possible samples, not predictions. Neither match settles the rules' expected ranking; the complete equally weighted outcomes can do that. **Hints, in order.** (1) "Could every offer in a round be the same?" (2) Build a word that forces score 6 whatever the rule does, then one forcing 0. (3) Use six copies for one match and reverse the two batches for the other.

Problem 3 / The best two-offer choices; core

First use the printed practice conversion on page 2: $(3, 1)$, take the first 3, leave the unseen 1 in the full record, score 3. Then distribute all nine $0/4/6$ cards across the table. Keep each card's full word; mark its one accepted score. Use the same first-offer decision on all cards beginning with that offer. Pool the cards before comparing totals.

Answer: pass 0, take 4, take 6; total 40, average $40/9$. Here is the full score key, grouped by the first offer:

First	Complete words	Scores	Take-first vs pass-first group totals
0	$(0, 0), (0, 4), (0, 6)$	$0, 4, 6$	0 versus 10: pass
4	$(4, 0), (4, 4), (4, 6)$	$4, 4, 4$	12 versus 10: take
6	$(6, 0), (6, 4), (6, 6)$	$6, 6, 6$	18 versus 10: take

Each first-offer group's choice is independent of the other groups. Maximizing all three totals proves the unique best first-offer rule. The scores are one 0, four 4s and four 6s. Fractions are unnecessary for this comparison. **Hints, in order.** (1) "Put cards with the same first offer together." (2) Try taking on all three, then passing on all three. (3) Compare the two totals in each group.

If the complete comparison is too much, work on one group and let another pair supply the other groups. Offer pages 6–7 directly when ready; upper pages are not a gate.

Concrete core / student pages 3, 6 and 7

Problem 4 / Two three-offer plans; core or a return visit

Score every card twice, on its A line and its B line. Split the 27 cards, including repeated scores and unseen tails, between children and pool a checked set; the adult may add. Here is every printed word with its scores; A/B denotes the two scores.

Word	A/B	Word	A/B	Word	A/B
$(0,0,0)$	0/0	$(4,0,0)$	4/0	$(6,0,0)$	6/6
$(0,0,4)$	4/4	$(4,0,4)$	4/4	$(6,0,4)$	6/6
$(0,0,6)$	6/6	$(4,0,6)$	4/6	$(6,0,6)$	6/6
$(0,4,0)$	4/4	$(4,4,0)$	4/4	$(6,4,0)$	6/6
$(0,4,4)$	4/4	$(4,4,4)$	4/4	$(6,4,4)$	6/6
$(0,4,6)$	4/4	$(4,4,6)$	4/4	$(6,4,6)$	6/6
$(0,6,0)$	6/6	$(4,6,0)$	4/6	$(6,6,0)$	6/6
$(0,6,4)$	6/6	$(4,6,4)$	4/6	$(6,6,4)$	6/6
$(0,6,6)$	6/6	$(4,6,6)$	4/6	$(6,6,6)$	6/6
Group total	40/40	Group total	36/40	Group total	54/54

B has the higher total: 134 **versus** 130, hence average $134/27$ versus $130/27$. A has one 0, thirteen 4s, thirteen 6s; B has two 0s, eight 4s, seventeen 6s. On $(4,0,0)$, A gains 4. On $(4,0,6)$ and the three $(4,6,*)$ words, B gains 2 each: net advantage 4. B does not beat A on every word. This comparison of two plans alone does not prove B globally optimal; Problem 6 supplies that explanation. **Hints, in order.** (1) "Show which offer A takes on this card, then which B takes." (2) Sort by the first offer to check that no card was lost. (3) Focus on first-4 cards if the plans' difference is hard to find.

Problem 7 / All best choices for new bags; core continuation

Pages 6–7 are available without pages 4–5. For each bag, compare taking and passing within the three cards for each first offer, as in Problem 3.

Bag	First 0: take/pass totals	Middle: take/pass totals	First 6: take/pass totals
$0,3,6$	$0/9$	$9/9$	$18/9$
$0,5,6$	$0/11$	$15/11$	$18/11$
$0,2,6$	$0/8$	$6/8$	$18/8$

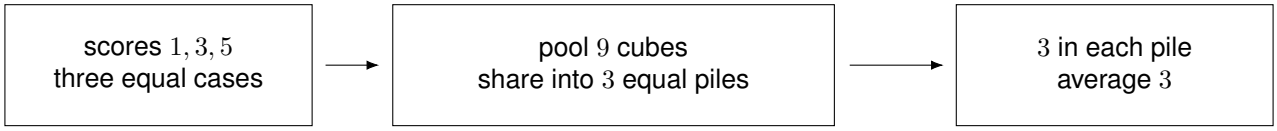
$0,3,6$**: exactly two best first-choice rules.** Both pass 0 and take 6; either take 3 or pass 3. Each totals 36, average 4. $0,5,6$**: uniquely pass** 0**, take** 5 **and take** 6**.** Total 44, average $44/9$. $0,2,6$**: uniquely pass** 0**, pass** 2 **and take** 6**.** Total 34, average $34/9$: here the middle ticket is passed, since $2 < 8/3$. Best scores on the printed cards, row by row:

Bag	First-0 row	Middle row	First-6 row
$0,3,6$	$0,3,6$	$3,3,3$ or $0,3,6$	$6,6,6$
$0,5,6$	$0,5,6$	$5,5,5$	$6,6,6$
$0,2,6$	$0,2,6$	$0,2,6$	$6,6,6$

Hints, in order. (1) "Try both choices on the middle-first cards." (2) Compare whole groups, not just the best-looking card. (3) Can two different choices have equal totals? Check all three first-offer groups.

Continuation / student pages 4–5

Readiness gate. Children need to interpret an average over equally likely outcomes and compare averages from sets of different sizes. If this is unfamiliar, use the following non-target example *before* Problem 5. Three possible scores $1, 3, 5$ have total 9. Share nine cubes equally among three places; each gets 3, so the average is 3. The cubes represent outcomes, not sequential offers.



The core remains available if this conversion needs more time. Fractions or adult-supported equal-sized batch comparisons are needed here; notation is optional, but averaging is part of the mathematics.

Problem 5 / A current 4 with two versus three offers left

Both diagrams count the current offer; their two and three counters are not numbers of *future* offers. Taking 4 gives average 4 in either position.

Offers left	Average if take	Best average if pass	Best choice
Two including this one	4	$(0+4+6)/3 = 10/3$	Take
Three including this one	4	$40/9$	Pass

With two left, passing forces the next offer: its three-case total is 10, versus 12 for three copies of a taken 4. With three left, passing leaves Problem 3's best two-offer rule and its nine-case total 40, versus 36 for nine copies of 4. The continuation includes another choice on the next offer. Do not replace its value with the next single draw's mean. **Hints, in order.** (1) "After you pass, how many counters will remain?" (2) Set out the complete remaining one- or two-offer cards. (3) Compare the continuation total with the same number of copies of score 4.

Problem 6 / The best three-offer plan, including remembered histories

Answer: no legal plan beats B. B first takes only 6; if continuing, on the second offer it takes 4 or 6 and passes 0; it takes the final offer. Total 134 on the 27 words, average $134/27$. The full score list is Plan B on guide page 5.

Object-based bound for every legal plan. Start at the end. The final offer is compulsory. After *any* passed history, the next tickets are still $0, 4, 6$ with equal chances. For a second offer 0, taking contributes 0 to its three possible final tails; passing contributes at most 10. At second 4, taking gives 12 versus passing 10; at 6, taking gives 18 versus 10. Consequently the best remaining nine-word total is $10 + 12 + 18 = 40$ after *each* first-offer history.

Now group the 27 words by the first offer. Taking first 0, 4 or 6 gives 0, 36 or 54 respectively on its nine tails. Passing permits at most 40 in each group. The group bounds are therefore $40, 40, 54$, total 134. Plan B attains all three bounds. A memory-dependent rule may choose differently after different first offers, but each group's future bag and bound are the same. A rule cannot use its remembered past to see an unseen tail. **Hints, in order.** (1) "What choices remain with one counter?" (2) Solve the last two offers separately after each possible passed first offer. (3) Compare the nine-word take and continuation totals in each first-offer group.

It is fine to keep a checked candidate plan as a conjecture and return to the bound on another visit. Recording "B beats A" is weaker than explaining "no legal plan can exceed 134".

Continuation / student page 8

Problem 8 / Three versus four offers from $0, 5, 6$

This requires fractional continuation averages or supported comparison of equivalent batches. Reuse the nine-card result from Problem 7, rather than asking children to transcribe 81 words. Let them reconstruct the later choices before revealing this answer table.

Offers	Optimal plan	Average
Three	Take 5 or 6 at either nonfinal offer; pass 0; final compulsory.	$143/27$
Four	First take only 6. After passing, use the three-offer plan above.	$448/81$

Here $V_1 = 11/3$, and Problem 7 gives $V_2 = 44/9$. With three offers including a current 5, taking wins because $5 > 44/9$. The best three-offer average is

$$V_3 = \frac{44/9 + 5 + 6}{3} = \frac{143}{27}.$$

With four offers including a current 5, passing wins because $143/27 > 5$. At a first 0, also pass; at a first 6, take. Thus

$$V_4 = \frac{143/27 + 143/27 + 6}{3} = \frac{448}{81}.$$

On all 27 complete three-offer words, the three-offer plan scores one 0, thirteen 5s and thirteen 6s: total 143. On all 81 four-offer words, the four-offer plan scores two 0s, 26 5s and 53 6s: total 448. The last three offers after either passed first 0 or passed first 5 each contribute 143 over their 27 tails; first 6 contributes $27 \cdot 6 = 162$. These are exact case totals, not a forecast for a short playing sample.

Hints, in order. (1) "What is your best remaining plan after passing the current 5?" (2) Use the new bag's two-offer total to value the remaining two offers. (3) Compare 5 with that continuation average; then repeat using three remaining offers.

Optional adult question: "Does adding time make a middling offer easier or harder to pass?" Do not generalize one changed choice to every distribution.

Problem 9 / Can one extra offer lower the optimum or leave it unchanged?

It cannot lower it. In an $(n+1)$-offer round, pass the first offer whatever it is, then use the old optimal n-offer plan. Independence and the unchanged bag reproduce its old average. The new optimum is at least as good as that legal plan. This argument works for any known finite bag of scores, including duplicate tickets or negative scores.

It can be unchanged: if every ticket has the same score c, every legal plan scores c, at every positive horizon. For example a bag $2, 2, 2$ has best average 2 with one, two or any number of offers.

For a nonconstant finite bag, each extra offer strictly improves it. Let the smallest score be a and largest M. The all-a word has positive probability and forces reward $a < M$ for every legal plan, while no reward can exceed M. Hence $V_n < M$. The next offer has positive probability of being M, so

$$V_{n+1} - V_n = \mathbb{E}[\max(X, V_n) - V_n] = \mathbb{E}[(X - V_n)_+] > 0.$$

In words: pass a new first offer if it is below the old average, but take it if above. Taking the largest score sometimes gives a strict improvement. The strict statement needs a finite horizon and a genuinely nonconstant bag. **Hints, in order.** (1) "Could you ignore the new first offer?" (2) Try a bag whose scores are all identical. (3) For a varied bag, can every legal plan still be forced to get the smallest score on some full word? Compare its average with the largest score.

Adult proof, sources and the next visit

Why the recurrence proves optimality. At one remaining offer the reward is forced, so the optimum before drawing is $\mathbb{E}[X]$. Suppose the best average for $m-1$ offers is V_{m-1} after every passed history. Independence makes that future problem the same after observing any current x. Taking gives x; passing allows at most V_{m-1}, attained by the earlier optimal plan. Choose the larger, then average over x. This proves the recurrence and an attainable bound for all legal histories by induction. A randomized choice is a weighted average of taking and passing, so cannot exceed their maximum. At equality any mixture is also optimal. Two printed student positions tie: the two-offer $0, 3, 6$ middle choice and a first 3 in the page 2 practice bag $1, 3, 5$.

Why complete words matter. With k equally likely tickets sampled independently, each length-n *ticket* word has probability k^{-n}. Stopping early leaves unseen tails in that collection. An accepted first offer represents k^{n-1} full words, not one case. Shortened stopped records have different weights; do not average a list with one representative per stopped record or per distinct score. With unequal probabilities, use product weights. Duplicate-score tickets can share a numerical word but retain their multiplicity.

Mathematical source. Thomas S. Ferguson, *Optimal Stopping and Applications*, Chapter 2, printed p. 2.1 (PDF p. 1), equation (1): general finite-horizon backward induction. Section 2.4, printed pp. 2.7–2.8 (PDF pp. 7–8), distinguishes sampling without replacement from iid offers; equation (6), p. 2.8, gives the common-distribution recursion used here. Section 2.1, p. 2.2 (PDF p. 2), gives the different relative-rank, best-choice secretary objective. Primary text: math.ucla.edu/ tom/Stopping/sr2.pdf. The finite-ticket instances, student tasks and guide examples are original applications of established mathematics.

Teaching lessons actually found, and our inferences.

- The organizer's *Pattern Block Exercises – Google Docs*, PDF p. 1, starts with five minutes of handling, then pairs/trios. The first-year report in the project README stresses concreteness and learning by doing. Our brief handling period and legal demonstration adapt that precedent.
- Givental, Nemirovskaya and Zakharevich, *Math Circle by the Bay* (AMS, 2018), Preface, printed pp. viii–x (PDF pp. 9–11). These pages discuss deep mathematical context, independent attempts, manipulatives, and changes in pace and depth. Our multiple-visit core/proof menu is an inference, not a session from the book.
- Rozhkovskaya, *Math Circles for Elementary School Students: Berkeley 2009 and Manhattan 2011* (AMS, 2014), Lesson 7, "At the lesson," items 1–2 (EPUB `OEBPS/part0017.xhtml`), reports likely missed legal chords and engaged K'nex polygon building. Lesson 8, item 2 (`part0018.xhtml`), reports copying and coloring delays and instructor help. Our partner-enforced deadline and supplied cards respond to those observations.

None of these sources prescribes this ticket activity, validates its physical implementation, or establishes its age fit. The adaptation is unpiloted. The current group record also reports a successful tiling session and a lamp rule children struggled to retain; visible state and partner checks are our response, not a tested remedy.

Returning children. Prior Weeks 24 and 42–46 include expected points, replacement draws or complete histories, but not this same finite-score acceptance task in the scoped research check. File presence is not evidence a child used or mastered them. Ask what was actually used. A later visit can finish the exact comparison, develop Problem 6's bound, or use Problems 8–9; preserve the accepted mathematical destination while pacing flexibly.

After the session, record the date, children and pages actually used; which handling rules required rescue; each fixed plan and one recoverable word or drawing worth keeping; whether a claim was *tried, conjectured, checked on all cases, or explained*; and what children wanted to continue. Separate observed behavior from hypotheses about its cause. Note actual prep/staffing and any failed physical pretest. Save unfinished complete sets and labelled totals for a return visit.